Knowledge Graph-Based Automated Short Answer Grading for Bengali

NLP Dominator

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Code: https://github.com/DigantaSen/Knowledge-Graphs-Based-Bengali-Answer-Evaluation
Models: https://huggingface.co/k040902/bengali_question_ans

Abstract

Automated short answer grading (ASAG) for Bengali is hampered by free word order, agglutinative morphology and the lack of labelled data and of a Bengali dependency parser. Similarity-only systems give near-perfect scores to answers whose roles are swapped (*Karak reversal*) or that are negated. We present a Bengali ASAG system that combines LaBSE similarity with (1) a predicate-centred knowledge graph compared by a Siamese graph attention network, (2) a Karak (semantic-role) validator, (3) negation, number and *answer-slot* features that compare only the part of an answer that answers the question, and (4) an XGBoost + MLP scorer trained with synthetic error examples. We evaluate on three datasets of increasing answer length with grouped splits, calibrated baselines and bootstrap confidence intervals. On normal answers the system reaches Pearson 0.83 / 0.95 / 0.91 (single / medium / long) against 0.63 / 0.92 / 0.79 for the best baseline, with significant MAE reductions on all three, and it scores hand-written role-swapped and negated answers at 0–18 instead of 50–90. We also report that the original datasets are heavily templated: a model trained on them alone scores natural medium-length answers worse than a baseline (Pearson 0.20 vs 0.74), which a small, rubric-scored clean synthetic dataset fixes.

1 Introduction

Automatic grading of short factual answers underpins intelligent tutoring and large-scale assessment. Progress has been strong for English (Mohler et al., 2011; Sultan et al., 2016; Riordan et al., 2017), but Bengali, spoken by over 230 million people, is harder: *free word order* lets subject, object and verb appear in almost any order; *fused grammatical markers* (case endings such as -কে, negation fused into verbs such as হয়নি) defeat exact token matching; and there is no production-quality dependency parser.

Multilingual sentence encoders such as LaBSE (Feng et al., 2022) and BanglaBERT (Bhattacharjee et al., 2022) are a strong baseline, but sentence similarity is insensitive to structure. A student who writes রাবণ রামকে বধ করেছিলেন ("Ravana killed Rama") when the reference is রাম রাবণকে বধ করেছিলেন shares every content word with it, yet is completely wrong. We call this failure mode *Karak reversal*, after the Sanskrit-derived concept of semantic roles (*karak*) in Bengali grammar. Negated answers (সূর্য পূর্বদিকে ওঠে না) and wrong numbers or places are similar blind spots. Our contributions are:

1. A Bengali ASAG pipeline with a predicate-centred knowledge graph and Siamese GAT, a Karak validator that handles case-marked objects and passives, sentence-aligned negation and number checks, and answer-slot features (§4).
2. An audit of the three datasets: duplicated pairs with conflicting scores, overlapping label bands and pervasive answer templates, with a grouped, deduplicated evaluation protocol (§3).
3. A rubric-scored clean synthetic dataset (514 rows, 54 new questions) and synthetic training errors, which expose and remove a template shortcut (§3.4).
4. An evaluation with calibrated baselines, bootstrap confidence intervals, unseen-question splits, an adversarial set and a component ablation.

2 Related Work

Automated short answer grading. Early ASAG combined lexical overlap with dependency alignment; Mohler et al. (2011) showed that dependency-tree alignment beats bag-of-words but relies on a parser that does not exist for Bengali. Sultan et al. (2016) used token alignment with paraphrase resources; neural models help under low resources (Riordan et al., 2017; Horbach et al., 2017). None of these distinguish a sentence from its Karak-reversed form when word order is free.

Models and Bengali tools. LaBSE (Feng et al., 2022) is trained contrastively on 109 languages; BanglaBERT (Bhattacharjee et al., 2022) is an ELECTRA discriminator. We found BanglaBERT mean-pooled word vectors unusable for word similarity (every pair of Bengali words has cosine ≥ 0.82), so LaBSE is used for words and graph nodes. Graph attention networks (Veličković et al., 2018) outperform GCNs (Kipf and Welling, 2017) on dependency-aware tasks (Zhang et al., 2019). For Bengali structure the tooling gap is severe: Stanza (Qi et al., 2020) has no Bengali model; Trankit (Nguyen et al., 2021) lists Bengali only as trainable, and the only Universal Dependencies treebank (BRU) has fewer than 1,000 units; the multilingual triple extractor mREBEL (Huguet Cabot et al., 2023) covers 18 languages without Bengali; IndIE (Mishra et al., 2023) was evaluated on Hindi; BanglaAutoKG (Wasi et al., 2024) builds corpus-level graphs with an LLM and untyped edges. We therefore build per-answer, typed graphs from BNLP POS tags with rules.

	Single	Medium	Long
Rows (raw)	1000	1000	320
Questions	50	50	16
Repeated pairs	331	508	29
After deduplication	669	492	291
Templated answers	53	695	113
Correct band	85–100	85–100	78–100
Partial band	35–80	40–75	25–77
Incorrect band	0–40	5–35	0–26
Train / test (normal)	568 / 101	418 / 74	247 / 44
Synthetic train rows	111	125	74

Table 1: Dataset statistics. Label bands are the observed score ranges per label. Every test set also contains the 51 adversarial samples.

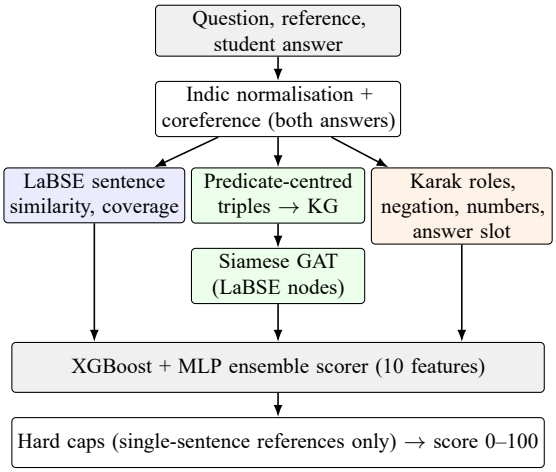


Figure 1: System architecture. Coreference is applied to both answers; three feature groups feed a learned ensemble, followed by hard caps for single-sentence references.

3 Data

3.1 Base datasets

Three datasets pair a question, a reference answer and a student answer with a score (0–100) and a label: *single-sentence* (1,000 rows), *medium* (1,000 rows, 2–3-sentence references) and *long* (320 rows, paragraphs). Our audit (Table 1) found four problems that affect evaluation:

- **Few questions**: 50 per dataset (16 for long), one reference each.
- **Duplicates**: 331 (single) and 508 (medium) rows repeat an earlier (reference, student) pair, and 193 of 200 repeated groups carry different scores. With a random row split, 47% (single) and 73% (medium) of test rows had an identical pair in training.
- **Overlapping label bands**: e.g. single-sentence *incorrect* answers reach 40 while *partially correct* starts at 35.
- **Templates**: a recurring 3-word prefix or suffix that never occurs in a reference (e.g. এই প্রশ্নের উত্তরে, আরও তথ্য প্রয়োজন।, আজ রবিবার।) appears in 53 single, **695** medium and 113 long student answers and largely determines the score.

We merge identical (question, reference, student) rows, averaging their scores, and split by identical pair (“unseen answers”) or by question (“unseen questions”).

3.2 Adversarial samples

We hand-wrote 51 adversarial samples, used for testing only: Karak reversal (17; বিড়াল ইঁদুর ধরেছে → ইঁদুর বিড়ালকে ধরেছে), negation (17), wrong number (4; 206 → 205 bones), wrong location (2), partial cover (1, scored 50) and voice change (10; the active reference rewritten as a passive with the roles reversed, পোরাসের দ্বারা আলেকজান্ডার পরাজিত হয়েছেন). All but partial cover are scored 0.

3.3 Synthetic training errors

Because adversarial samples never appear in training, a scorer cannot learn that such answers score 0. We therefore add rule-generated errors to the *training split only*: each reference sentence is negated (না after the final verb, নয় after a nominal predicate), its subject and -কে object are swapped, or its first number is changed by one, and an unchanged copy is added so that short pairs are not all wrong. Synthetic rows are capped at 30% of the real training rows, and sentences used by the adversarial samples are excluded.

3.4 Clean synthetic dataset

To test whether the system grades natural answers rather than templates, we wrote a second dataset of 514 rows for 54 new questions (30 single, 16 medium with three key facts each, 8 long with five). Each student answer has a declared type and its score comes from a fixed rubric, not human graders: for single sentences, copy 100, paraphrase 95, correct passive or other construction 92, extra correct detail 92, brief key answer 88, partial 55, key right but a detail wrong 40, wrong entity 10, wrong detail 8, and role swap, negation, off-topic or “don’t know” 0. For medium and long answers each key fact is worth $95/k$ points and a contradicted fact costs a further 10. Labels use non-overlapping bands ($\geq$85 correct, 35–84 partial). The rows are marked `source=clean_synthetic`; we report them separately.

4 Method

Figure 1 shows the pipeline.

Normalisation and coreference. Text is normalised with the Indic NLP library. A rule-based resolver replaces whole-token pronouns (তিনি, এটি, …) with the most recent proper noun (persons) or common noun (things), seeded from the reference first because the question contains the interrogative, not the answer. It is applied to both answers so that a verbatim copy stays a copy.

Roles from POS tags. BNLP’s tagger emits the BIS tagset (`NC`, `NP`, `VM`, …), mapped to universal POS. A rule

parser labels the first unmarked nominal as subject, -কে-marked nominals as explicit objects, other nominals after the subject as unmarked objects, the nominal before দ্বারা/কর্তৃক as the agent of a passive (whose grammatical subject becomes the patient), and genitive and locative nominals by explicit endings. Lemmas strip case endings (রামকে → রাম).

Predicate-centred knowledge graph. Each clause becomes *subject* $\xrightarrow{\text{agent}}$ PREDICATE $\xrightarrow{\text{object/location/time/count}}$ *argument*, with genitives attached to the noun they modify and adjectives as qualities. The predicate node carries the verb, the noun of a compound verb (বধ করেছিলেন) and any negation (ওঠে না); proper-noun runs and number+unit are merged into single nodes. Node features are the LaBSE embedding of the node text plus a one-hot node type and a negation flag (775 dimensions); edge features are the weight, a one-hot relation type and a direction flag, with reverse edges added.

Siamese GAT. Two GATv1 layers (4 heads, 256 → 128 dimensions, edge features of dimension 9) with layer normalisation and mean pooling embed each graph; the cosine of the two embeddings is regressed onto score/100 (MSE, Adam, lr 0.005, 5 epochs, batch 32, seeded).

Karak validator. Roles are compared per aligned sentence pair (sentences paired by token overlap). A swap is counted only when the student marks the moved entity explicitly (-কে or passive patient): a reference agent that appears as a marked object, or a reference object that becomes the subject while the subject changes, adds 0.5. Identical inflected words in another order are a legal reordering and exempt. Missing locations, numbers and modifiers and a negation mismatch add smaller penalties (capped at 1).

Negation, numbers and entities. Negation is detected on punctuation-stripped tokens, including fused -নি forms, and compared per aligned sentence pair. A wrong number is a digit run in the student answer that the reference lacks. Entity mismatch is the share of reference content words (plus NER entities) missing from the student answer, with LaBSE similarity ≥ 0.85 counting as a synonym (measured: synonyms 0.84–0.98, different entities 0.31–0.83).

Answer-slot features. Words that also occur in the question are shared by right and wrong answers alike. Removing them leaves the *answer content*, from which we compute its recall and precision between reference and student, and *key_match*: whether the key answer item is present (the number for কবে/কত, the place or direction for কোথায়, the item named by কোন X, else the first proper noun, skipping titles such as সম্রাট).

Scorer and overrides. Ten features (LaBSE similarity, graph similarity, Karak penalty, entity mismatch, sentence coverage, negation and wrong-number flags, and the three answer-slot features) plus twelve products and differences feed XGBoost (300 trees, depth 5) and an MLP (128-64-32). The scorer is trained on *out-of-fold* GNN similarities so that it sees unseen-row values, and the XGBoost/MLP weight is chosen on out-of-fold training predictions. For single-sentence references only, a negation mismatch caps the score at 30 and a wrong number at 25 (fixed rules); multi-sentence answers receive partial credit from human graders, so there the flags are left to the scorer.

5 Experimental Setup

Splits hold out 15% of groups (identical pairs, or questions) with seed 42; the 51 adversarial samples are always test-only. Baselines map a single similarity to the score scale with a linear regression fitted on the training split: LaBSE cosine, ROUGE-L (with a Bengali tokenizer), LaBSE + ROUGE-L, BanglaBERT cosine (with its official normaliser) and the GNN alone. We report Pearson correlation, MAE, the share of answers within ± 10 points, and binary F1 of the pass class (score ≥ 50); 95% confidence intervals and p-values come from a paired bootstrap (2,000 resamples) of the proposed system against the best baseline (lowest MAE). Cross-validation uses five grouped folds over all normal rows, with a fresh GNN and scorer per fold. Everything runs on a CPU; a full run of all three datasets with cross-validation takes about 25 minutes.

6 Results

Main results. Table 2 gives the full test sets and Table 3 the normal answers. The proposed system beats every calibrated baseline on all three datasets; on normal answers the improvement over the best baseline is significant for both Pearson and MAE on all three. Cross-validation agrees with the held-out test. The margin is smallest on medium, where calibrated ROUGE-L already reaches 0.92 because templates make surface overlap predictive. On non-templated normal answers the gain on single-sentence answers holds (Pearson 0.789 vs 0.511, ΔMAE -9.2, $p < 0.001$) and is significant on medium (0.885 vs 0.651, $n = 18$), while long ($n = 29$) is not significant.

Unseen questions. Table 4 holds out whole questions, so test references never appear in training. The system keeps its advantage: MAE falls by 4.7–7.7 points on all three datasets ($p < 0.001$), and Pearson is significantly higher on single and medium answers.

Adversarial answers. Table 5 breaks the adversarial set down. Role swaps, negation, wrong numbers and voice change are scored 0–32 (true score 0), mostly learned by the scorer from the synthetic training errors: the hard caps fire on at most one answer per dataset. Wrong location (not synthesised) remains the weakest category, especially for medium and long answers.

Generalisation to non-templated answers. Table 6 trains on the original data with and without the clean synthetic rows and tests on unseen questions; the "clean" rows are clean-dataset answers to questions neither model saw. Trained on the original data alone, the model is *worse than a*

Model	Single-sentence (n=152) Pearson	MAE	F1	Medium (n=125) Pearson	MAE	F1	Long (n=95) Pearson	MAE	F1
LaBSE cosine	0.306	30.26	55.0	0.028	32.68	50.8	0.108	35.57	37.4
ROUGE-L	0.514	26.14	54.9	0.279	27.73	54.7	0.234	31.78	46.2
LaBSE + ROUGE-L	0.436	28.35	61.7	0.246	28.38	55.7	0.133	34.79	40.4
BanglaBERT cosine	0.241	32.00	48.4	0.225	30.10	54.2	0.296	32.67	56.7
GNN only	0.442	27.10	52.9	0.308	28.47	63.7	0.347	32.21	48.8
Proposed	**0.830**	**12.10**	**80.4**	**0.855**	**9.49**	**86.0**	**0.788**	**12.19**	**72.3**

Table 2: Full test sets (normal + 51 adversarial rows), unseen-answer split. Baselines are linearly calibrated on the training split. The pooled numbers mix two populations; Table 3 reports normal answers alone.

Dataset (n)	Best baseline	Baseline P / MAE	Proposed Pearson [95% CI]	Proposed MAE [95% CI]	ΔPearson (p)	ΔMAE (p)	CV Pearson
Single (101)	LaBSE + ROUGE-L	0.626 / 21.21	0.829 [0.743, 0.895]	12.71 [10.30, 15.35]	+0.203 (<0.001)	−8.49 (<0.001)	0.801 ± 0.030
Medium (74)	ROUGE-L	0.922 / 11.52	0.951 [0.930, 0.967]	6.79 [5.58, 8.04]	+0.028 (0.013)	−4.72 (<0.001)	0.944 ± 0.012
Long (44)	LaBSE + ROUGE-L	0.793 / 13.46	0.907 [0.822, 0.959]	8.42 [6.06, 11.15]	+0.114 (0.019)	−5.04 (0.020)	0.916 ± 0.036

Table 3: Normal (non-adversarial) test answers, unseen-answer split, with paired-bootstrap 95% confidence intervals against the best baseline, and five-fold grouped cross-validation.

Dataset (n)	Best baseline P / MAE	Proposed P / MAE	ΔMAE (p)
Single (100)	0.607 / 21.49	**0.829** / **13.76**	−7.73 (<0.001)
Medium (82)	0.896 / 13.47	**0.924** / **8.72**	−4.75 (<0.001)
Long (54)	0.863 / 14.34	**0.916** / **8.57**	−5.77 (<0.001)

Table 4: Normal answers to unseen questions (split grouped by question). ΔPearson: +0.222 ($p < 0.001$), +0.028 ($p = 0.028$), +0.053 ($p = 0.16$).

Category (n)	Single	Medium	Long
Karak reversal (17)	14.9	18.4	15.4
Negation (17)	0.7	0.3	5.8
Wrong number (4)	4.5	6.4	4.2
Wrong location (2)	32.4	62.8	45.4
Partial cover (1, true 50)	34.2	54.2	52.3
Voice change (10)	19.1	21.2	31.8

Table 5: Mean predicted score per adversarial category (true score 0, partial cover 50).



Figure 2: Predicted vs. true score, single-sentence test set: normal answers (blue) and adversarial answers (red).

baseline on natural medium answers (Pearson 0.20 vs 0.74) and long answers (0.60 vs 0.91): it had partly learned the templates. Adding the clean rows and the answer-slot features raises it to 0.74 / 0.86 / 0.95. Table 7 shows the remaining errors by answer type: differently worded correct answers are still under-scored and wrong-entity answers over-scored, though both moved in the right direction.

Feature importance. The strongest XGBoost feature is `max_pen`, the larger of the Karak penalty and the entity mismatch (0.34 single, 0.68 medium, 0.29 long), followed by the negation flag on single sentences (0.24) and the Karak penalty on long answers (0.28). The answer-slot features rank among the top five on single-sentence answers. Graph similarity is not a top feature by importance, and the GNN alone is the weakest single signal, yet removing it costs 0.055 Pearson and 1.8 MAE points (Appendix A): it is complementary to the sentence-level features rather than a replacement.

7 Analysis

Case studies. We score five answers with the trained single-sentence model. **(1) Verb omitted.** গীতাঞ্জলি কে লিখেছিলেন? Reference রবীন্দ্রনাথ ঠাকুর গীতাঞ্জলি লিখেছিলেন।, student রবীন্দ্রনাথ ঠাকুর গীতাঞ্জলি।: 63.4. The key answer is present (*key_match*=1, LaBSE 0.85) but the answer is still under-scored, the remaining weakness of Table 7. **(2) Karak reversal.** আলেকজান্ডার পোরাসকে পরাজিত করেছেন। vs. পোরাস আলেকজান্ডারকে পরাজিত করেছেন।: LaBSE similarity is 0.98, but the validator detects both swaps (penalty 1.0) and the answer scores 7.1. **(3) Wrong number.** 206 vs. ২০৭টি bones: 21.6, from the wrong-number penalty and a missing key, without needing the cap. **(4, 5) Negation.** সূর্য পূর্বদিকে ওঠে না। and ক্রিস্টোফার কলম্বাস আমেরিকা আবিষ্কার করেননি। (fused negation) both score 0.0; the previous version of the sys-

Tier (rows)	Baseline	Original	+clean	+slot
Single (66)	0.41	0.57	0.65	**0.74**
Medium (24)	0.74	0.20	0.78	**0.86**
Long (14)	0.91	0.60	0.87	**0.95**

Table 6: Pearson on clean-dataset answers to unseen questions: best baseline, model trained on original data only, plus clean rows, plus answer-slot features.

Answer type	Rubric	+clean	+slot
Paraphrase	95	54	67
Brief (শাহজাহান।)	88	45	62
Correct passive	92	58	63
Wrong entity	10	42	32
Wrong detail	8	43	38

Table 7: Mean predicted score per answer type, single-sentence clean answers to unseen questions (6 rows per type).

tem scored them 72.5 and 73.3.

What did not work. Up-weighting the clean rows three times lowered performance on both clean and original rows. Generating brief and wrong-entity training answers by rule produced about two-thirds mislabelled rows, because the tagger marks many common words as proper nouns, so it was dropped.

Earlier version of this report. An earlier version reported higher cross-validation scores (Pearson 0.87–0.93) and lower adversarial performance. The former came from duplicated pairs crossing the train/test boundary and from a GNN warm-started on the cross-validation rows; the latter from bugs that disabled the Karak validator, negation detection and ROUGE-L for Bengali. All numbers here use the corrected code.

8 Limitations and Future Work

Data. The original datasets are synthetic and templated, with duplicated pairs and overlapping labels; the clean dataset is rubric-scored by its authors, not by independent graders, and not yet reviewed by a native speaker. Test sets are small (44–101 normal answers), so differences on the non-templated medium and long subsets are not significant. The 51 adversarial samples were inspected during development; a fresh set written by someone who has not seen the rules would give a cleaner estimate, and synthetic training errors share their perturbation types with it.

Remaining errors. Differently worded correct answers are still under-scored and wrong-entity answers over-scored (Table 7); correct passive paraphrases also score low; wrong locations are poorly handled. The rule-based coreference resolver did not improve results on these datasets (Appendix A).

Parser bottleneck. Roles come from POS tags and case endings, so tagger errors still choose the wrong subject. A Bengali dependency treebank and parser, or LLM-based triple extraction, are the highest-impact next steps; human-graded, non-templated answers are the most valuable data.

9 Conclusion

We presented a Bengali ASAG system that combines LaBSE similarity with a predicate-centred knowledge graph, a Karak validator that handles case marking and passives, sentence-aligned negation and number checks, and answer-slot features, trained with synthetic error examples. With a leakage-free, grouped evaluation it significantly outperforms calibrated baselines on all three datasets (Pearson 0.83 / 0.95 / 0.91 on normal answers) and scores role-swapped and negated answers close to zero. The audit also shows that the original datasets reward templates, and that a small clean dataset is needed to grade natural answers reliably.

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A Component Ablation

Table 8 removes one component at a time on the single-sentence dataset (unseen-answer split). Features are extracted once, the component's features are neutralised and the scorer is retrained; the full system here is a separate training run, so it differs slightly from Table 3 (run-to-run variation of about 0.02 Pearson).

Variant	Normal answers Pearson	MAE	F1	Adv. MAE
Full system	0.845	11.89	84.4	10.55
– GNN branch	0.790	13.67	80.0	11.62
– Karak validator	0.812	14.34	75.3	31.47
– Entity matching	0.804	14.39	80.8	8.33
– Answer-slot features	0.791	14.44	81.0	12.07
– Negation / number flags	0.846	11.91	83.5	10.17
– Hard overrides	0.845	11.89	84.4	10.55
– Synthetic training errors	0.812	13.43	83.5	40.26
– Coreference	0.863	10.17	86.4	12.07

Table 8: Leave-one-component-out ablation, single-sentence test set (101 normal answers, 51 adversarial).

The GNN, Karak validator, entity matching and answer-slot features each lower MAE on normal answers by 1.8–2.6 points. For adversarial answers the synthetic training errors (MAE 10.6 → 40.3 without them) and the Karak validator (31.5) are essential. The negation and number flags and the hard overrides change little, because the Karak penalty already includes the negation and number checks and the scorer learns these cases. Coreference resolution does not help on this data: without it normal answers score slightly better, while adversarial MAE rises from 10.6 to 12.1.

B Confusion Matrix and Adversarial Breakdown

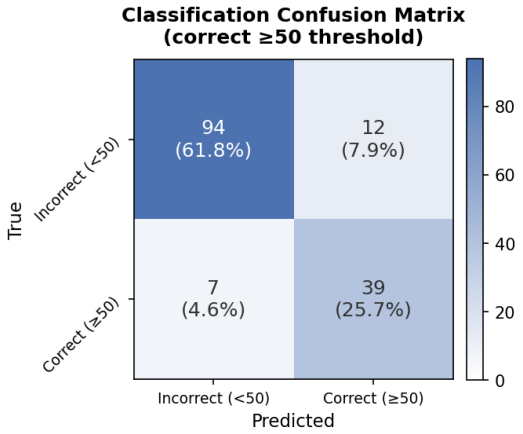


Figure 3: Pass/fail confusion matrix (score ≥ 50), single-sentence test set.

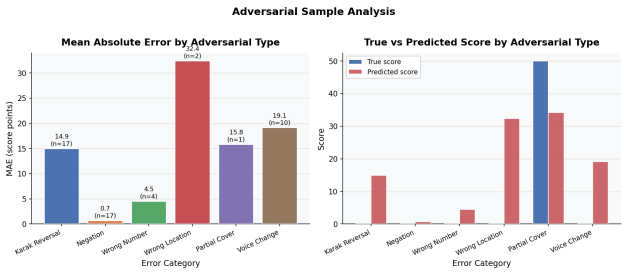


Figure 4: Mean absolute error and mean predicted score per adversarial category, single-sentence test set.